Paper Radio: Synthetic Forecasting Study

SYNTHETIC DEMO - NOT PUBLISHED RESEARCH

Abstract
This fictional document demonstrates Paper Radio's source retrieval. All datasets, experiments, and numerical results below are invented for software testing.

Research question
Can a simple seasonal forecasting model improve a synthetic daily demand baseline?

Methods
The synthetic dataset contains 365 daily observations. The baseline predicts the previous day's demand. A seasonal model adds seven-day lag features. The evaluation uses three chronological folds with a seven-day forecast horizon. Earlier observations are used for training; future observations are held out for evaluation.

Evaluation and results

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Evaluation

The metric is mean absolute error (MAE), measured on each chronological test fold. The baseline has synthetic MAE values of 12, 11, and 13 units. The seasonal model has synthetic MAE values of 9, 10, and 8 units. These values are invented test fixtures, not experimental findings.

Reporting

The fictional results suggest lower error for the seasonal model in these three synthetic folds. No significance test, confidence interval, production trial, or financial impact measurement is reported. No claim about real customer demand follows from this demonstration.

Limitations and next steps

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Limitations

The study uses synthetic data only. It does not establish that the forecasting model will generalize to real demand. The dataset covers one simulated year, and the evaluation uses only three folds. Promotions, stockouts, missing observations, and changes in customer behavior are not modeled.

Next steps

Evaluate on a real public dataset, compare against stronger baselines, and examine error across seasonal periods before proposing deployment.

Questions this demo cannot answer

This document does not report training cost, implementation language, model parameter count, production latency, or measured business value.